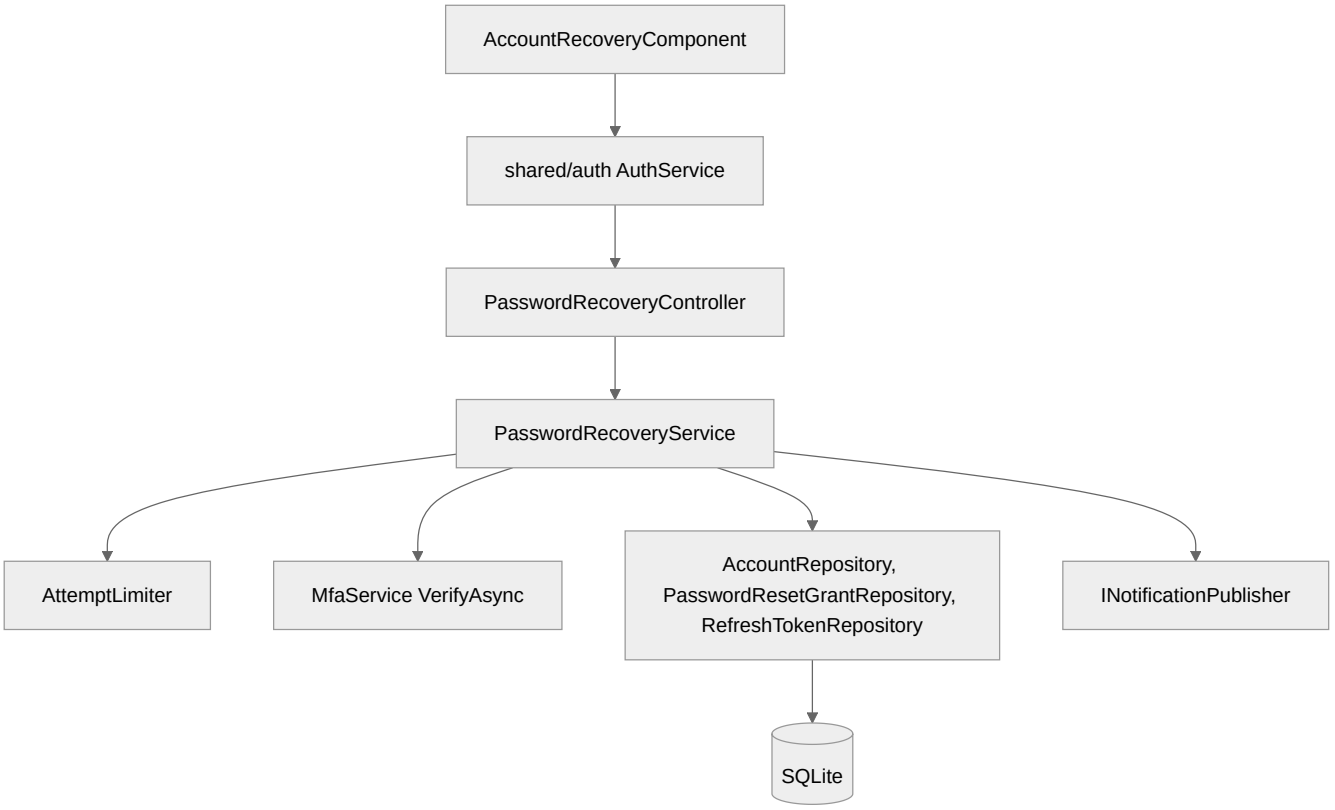
Account Recovery – Technical Documentation

Project: TaskFlow – Intelligent Task Allocation & Tracking System
Parent Module: Authentication · **Child Module:** 01.2 Account Recovery
Version: 1.0 · **Last Updated:** 31-Oct-2026
Backend: ASP.NET Core Web API (.NET 10) · **Frontend:** Angular · **Database:** SQLite (EF Core)
Author: Jojo George (jojogeorge3344@gmail.com)
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1. Technical Overview

Account Recovery (password reset) is a two-call anonymous flow. `PasswordRecoveryController` → `PasswordRecoveryService.VerifyAuthenticatorAsync` accepts an account identifier and a current authenticator code (no backup code) and, after the in-memory `AttemptLimiter` allows it, returns a single-use, 10-minute reset token whose SHA-256 hash is stored in `PasswordResetGrants`. `ResetPasswordAsync` then sets the new password, rotates the security stamp, revokes all refresh tokens, cancels the user's other grants and publishes a System notification. On the client, `AccountRecoveryComponent` runs the two steps (`RecoveryStep` = `verify` / `password`).

2. Architecture Flow



3. Backend File Structure

All paths are under `TaskFlow Proj/backend/src/TaskFlow.Api/`.

Layer	Path	Files
Controller	`Controllers/Auth/`	`PasswordRecoveryController.cs`
Service	`Services/Auth/PasswordRecovery/`	`IPasswordRecoveryService.cs`, `PasswordRecoveryService.cs`
Service (used)	`Services/Auth/AttemptLimits/`	`IAttemptLimiter.cs`, `AttemptLimiter.cs` (shared, see parent section 7)
Service (used)	`Services/Auth/Mfa/`	`IMfaService.cs`, `MfaService.cs` (`VerifyAsync`, see 01.3)
Repository	`Repositories/Auth/PasswordRecovery/`	`IPasswordResetGrantRepository.cs`, `PasswordResetGrantRepository.cs`
Repository (used)	`Repositories/Auth/Accounts/`, `Repositories/Auth/RefreshTokens/`	`AccountRepository.cs`, `RefreshTokenRepository.cs`
DTOs	`Dtos/Auth/PasswordRecovery/`	`PasswordRecoveryRequests.cs` (`AuthenticatorRecoveryRequest`, `RecoverPasswordRequest`, `RecoveryTokenResponse`, `RecoveryVerifyStatus`, `RecoveryVerifyResult`)
Validators	`Dtos/Auth/PasswordRecovery/Validators/`	`PasswordRecoveryValidators.cs` (`AuthenticatorRecoveryRequestValidator`, `RecoverPasswordRequestValidator`)
Entities	`Models/Entities/`	`PasswordResetGrant.cs`, `User.cs`, `RefreshToken.cs`
Enums	`Dtos/Auth/PasswordRecovery/PasswordRecoveryRequests.cs`, `Models/Enums/`	`RecoveryVerifyStatus` (Verified, Failed, Throttled); `NotificationType` (System)

4. Frontend File Structure

All paths are under `TaskFlow Proj/frontend/src/app/`.

Type	Path	Purpose
Routes	`modules/auth/auth.routes.ts`	`account-recovery` route (outside the shell)
Page component	`modules/auth/account-recovery/pages/account-recovery.component.ts`	Two-step recovery (`RecoveryStep` = `verify` / `password`)
Model	`modules/auth/account-recovery/pages/account-recovery.model.ts`	`RecoveryStep`

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Type	Path	Purpose
API service	`shared/auth/auth.service.ts`	`verifyAuthenticatorRecovery`, `recoverPassword`
Models	`shared/auth/auth.models.ts`	`RecoveryTokenResponse`, `RecoverPasswordRequest`
Messages	`shared/auth/login-reasons.ts`	Login notice after a successful reset

5. API & Logic

5.1 API Endpoints

#	Method	Endpoint	Description	Permission (Policy)	Request DTO	Response DTO
4	POST	`/api/v1/auth/password-recovery/authenticator/verify`	Step 1: account identifier + current authenticator code → single-use reset token	`[AllowAnonymous]`	`AuthenticatorRecoveryRequest`	`RecoveryTokenResponse` / `400` / `429` (ProblemDetails)
5	POST	`/api/v1/auth/password-recovery/reset`	Step 2: reset token + new password	`[AllowAnonymous]`	`RecoverPasswordRequest`	`204 No Content` / `400` (ProblemDetails or validation problem)

5.2 Request / Response Samples

```
// POST /api/v1/auth/password-recovery/authenticator/verify
{ "identifier": "jane", "code": "654321" }
```

```
// 200 OK
{ "resetToken": "q3V0b2tlbl9leGFtcGxlX2Jhc2U2NHVybA" }
```

```
// POST /api/v1/auth/password-recovery/reset
{ "resetToken": "q3V0b2tlbl9leGFtcGxlX2Jhc2U2NHVybA", "newPassword": "NewPass1",
"confirmPassword": "NewPass1" }
```

```
// 400 Bad Request – token unknown, used or expired
{ "status": 400, "detail": "This reset link is no longer valid. Start again from the login
page." }
```

5.3 Service Layer Logic

Method: `PasswordRecoveryService.VerifyAuthenticatorAsync(string identifier, string? code, string? remoteAddress, CancellationToken)`

1. Trims the identifier. `IAttemptLimiter.IsAddressBlocked(remoteAddress)` → `Throttled` before any database work.
2. `IAccountRepository.FindActiveByLoginForUpdateAsync(identifier)` (active and not deleted).

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3. Budget key: `IAttemptLimiter.AccountKey(user.Id)` (`user:{id}`) or `UnknownKey(identifier)` (`name:{UPPER}`) for unknown names. `TryAcquire(accountKey, remoteAddress)` reserves an attempt **before** checking the code; refused → `Throttled`.
4. User null or `MfaEnabledAt` null → `Failed`.
5. `IMfaService.VerifyAsync(user, code, allowBackupCode: false, countFailure: false)` – a wrong code does not touch the account's MFA failure counter; an existing MFA lock still refuses.
6. On success: `_limiter.Reset(accountKey)` and `IssueResetTokenAsync(user.Id)` → `Verified` with the token.
7. Controller maps `Verified` → `200`, `Throttled` → `429` ("Too many attempts. Try again in a few minutes."), otherwise `400` ("The details are incorrect, or the code has expired. Check them and try again."). The caller address comes from `HttpContext.Connection.RemoteIpAddress`.

Method: `PasswordRecoveryService.IssueResetTokenAsync(int userId)` – 32 random bytes, Base64Url without padding; stores a `PasswordResetGrant` with `TokenHash` = SHA-256 hex, `ExpiresAt = now + 10 minutes` (`TokenLifetime`).

Method: `PasswordRecoveryService.ResetPasswordAsync(string resetToken, string newPassword)`

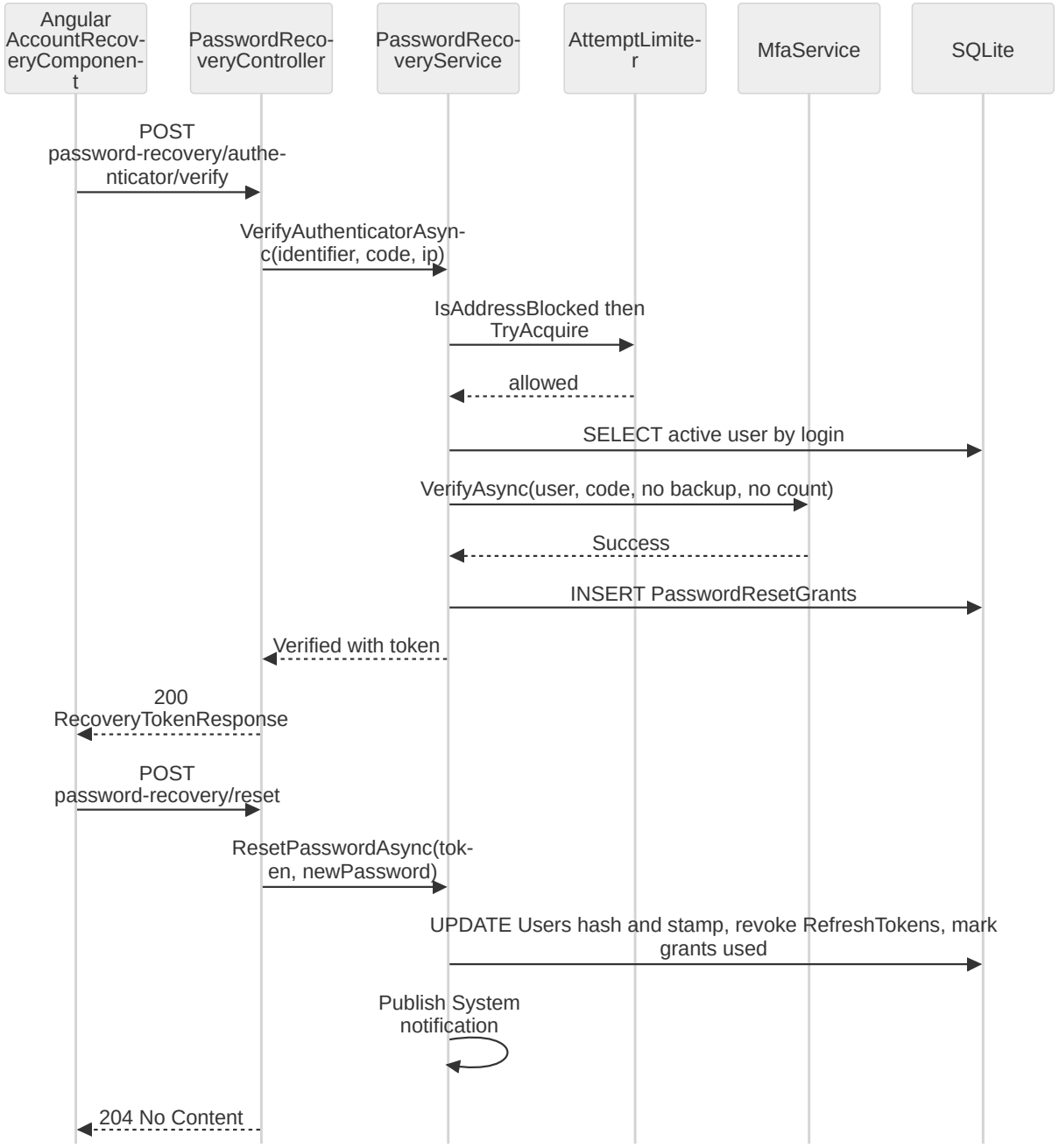
1. `FindUnusedByHashWithUserForUpdateAsync(hash)`; false if no grant, no user, expired, user deleted or inactive.
2. Hashes the new password, rotates `SecurityStamp` (`Guid.NewGuid().ToString("N")`), sets `UpdatedAt` and `grant.UsedAt`.
3. Revokes all active refresh tokens (`GetActiveForUsersForUpdateAsync`) and marks all other unused grants used (`GetOtherUnusedForUpdateAsync`).
4. `SaveChangesAsync`; a `ConcurrencyConflictException` (token spent in parallel; `UsedAt` is a concurrency token) → false.
5. Publishes a `NotificationType.System` notification: "Your password was reset with your authenticator app. If this was not you, contact an administrator now." Failures are logged as warnings and not returned to the caller.

5.4 Repository Layer

Method	Description	Query Notes
`AccountRepository.FindActiveByLoginForUpdateAsync(identifier)`	Active, not deleted user by username or email	Tracked, no role include
`PasswordResetGrantRepository.FindUnusedByHashWithUserForUpdateAsync(tokenHash)`	Unused grant by hash, with user	Tracked; `Include(User)`; `UsedAt == null`
`PasswordResetGrantRepository.GetOtherUnusedForUpdateAsync(userId, exceptGrantId)`	User's other unused grants	Tracked
`RefreshTokenRepository.GetActiveForUsersForUpdateAsync(userIds)`	Unrevoked tokens of the users	Tracked; `RevokedAt == null`

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5.5 Sequence Diagram



6. Database Design

This child owns `PasswordResetGrants`. Full `Users` column list and relationships: parent document section 6; `RefreshTokens` columns: 01.4.

Table	Column	Type	Constraints	Description
`PasswordResetGrants`	`Id`	INTEGER	PK	–

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Table	Column	Type	Constraints	Description
`PasswordResetGrants`	`UserId`	INTEGER	FK → `Users.Id`; index `IX_PasswordResetGrants_UserId_UsedAt`	Owner
`PasswordResetGrants`	`TokenHash`	TEXT(64)	Not Null; index `IX_PasswordResetGrants_TokenHash`	SHA-256 hex of the reset token
`PasswordResetGrants`	`CreatedAt`, `ExpiresAt`	TEXT (DateTime)	Not Null	Lifetime 10 minutes
`PasswordResetGrants`	`UsedAt`	TEXT (DateTime)	Nullable, concurrency token	Spent or cancelled
`Users`	`Username`, `Email`, `IsActive`, `IsDeleted`	TEXT / INTEGER (bool)	See parent	Identify the active account
`Users`	`MfaEnabledAt`, `MfaSecretProtected`, `MfaLastUsedStep`, `MfaLockedUntil`	TEXT / INTEGER	See parent	Recovery needs MFA on; code verified against the key; existing lock refuses
`Users`	`PasswordHash`, `SecurityStamp`, `UpdatedAt`	TEXT	Not Null	New hash, rotated stamp on reset
`RefreshTokens`	`RevokedAt`	TEXT (DateTime)	Nullable	Set for every active token of the user on reset

Relationships

- `PasswordResetGrants.UserId → Users.Id` (FK, Cascade)

7. Security & Authorization

- Both endpoints are `[AllowAnonymous]`; possession of the authenticator (TOTP code) is the only recovery factor; backup codes are refused.
- **Anti-enumeration:** one message for every failure (unknown account, inactive, MFA off, wrong code).
- **Throttling:** `AttemptLimiter` – 5 attempts per account key and 20 per caller address in a sliding 5-minute window; the attempt is reserved before the code is checked (limiter details in the parent document).
- **Lockout isolation:** `countFailure: false`, so a stranger cannot lock the owner's sign-in codes; an existing MFA lock still refuses.
- **Reset token:** 32 random bytes, stored only as SHA-256 hex, 10 minutes, single use (`UsedAt` concurrency token); all other unused grants are cancelled when one is used.
- **Session effects:** reset rotates `SecurityStamp` and revokes all refresh tokens, so every session ends (validated in 01.4).

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Permission Key	Endpoint(s)	UI Element Controlled
(none – `[AllowAnonymous]`)	`POST /api/v1/auth/password-recovery/authenticator/verify`, `POST /api/v1/auth/password-recovery/reset`	Account Recovery screen (both steps)

8. Validations & Error Handling

Rule	Layer	HTTP Code	Error Message
`Identifier` required, max 256; `Code` required, max 32	`AuthenticatorRecoveryRequestValidator`	400	FluentValidation defaults
Recovery failure (any reason)	`PasswordRecoveryController`	400	The details are incorrect, or the code has expired. Check them and try again.
Recovery throttled	`PasswordRecoveryController`	429	Too many attempts. Try again in a few minutes.
`ResetToken` required; `NewPassword` 6–128; `ConfirmPassword` equal	`RecoverPasswordRequestValidator`	400	Passwords do not match. (others FluentValidation defaults)
Reset token unknown, used, expired, user inactive	`PasswordRecoveryController.Reset`	400	This reset link is no longer valid. Start again from the login page.

9. Integrations / Side Effects

- **Notifications:** `INotificationPublisher.PublishAsync(..., NotificationType.System, ...)` – password recovery reset → "Your password was reset with your authenticator app. If this was not you, contact an administrator now."
- **Email:** none. No password, code or link is sent by email or SMS.
- **Activity Log:** none; recovery notification failures are logged as warnings.
- **Session effects:** `SecurityStamp` rotated and all refresh tokens revoked (all sessions end).

10. Configuration

Not applicable: this child has no `appsettings.json` keys of its own.

Hard-coded values: `PasswordRecoveryService.TokenLifetime` = 10 min. The `AttemptLimiter` limits are listed in the parent document.

11. Testing

Paths are relative to `backend/tests`.

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Test Type	File / Location	Covers
Unit	`TaskFlow.UnitTests/Auth/PasswordRecovery/PasswordRecoveryServiceTests.cs`	Recovery verify and reset
Integration	`TaskFlow.IntegrationTests/Auth/Mfa/MfaRecoveryEndpointsTests.cs`	Recovery with authenticator
Integration	`TaskFlow.IntegrationTests/Auth/PasswordRecovery/PasswordRecoveryEndpointsTests.cs`	Recovery endpoints

Shared: `AttemptLimiterTests.cs` is listed in the parent document. No frontend spec files were found under `modules/auth/account-recovery`.

12. Known Limitations / Gaps found

- Users without MFA (possible only when `Mfa:Required` is false) cannot use self-service recovery.
- **Recovery note text:** the Account Recovery screen says "turned on two-step verification from your Profile", but setup happens on `/security-setup`; the permission audit notes there is no MFA card on Profile.
- **Stale frontend comments:** `auth.models.ts` describes `RecoveryTokenResponse` / `RecoverPasswordRequest` as the "SMS" reset, but recovery uses the authenticator code only.

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